Xiaoya Li

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EDUCATION

New York University **Brooklyn, NY**
- M.S. in Computer Engineering Expected May 2027
- GPA: 4.0/4.0

Anhui University **Hefei, China**
- B.S. in Internet Banking Graduated June 2025
- GPA: 3.64/4.0

Relevant Coursework: Optimal Transport, Embedded System, Computer System Architecture, Big Data Analysis and Data Mining, Machine Learning, Python Programming

RESEARCH EXPERIENCE

Control, Robotics, and Embodied Intelligence (CREO) Lab **New York University, NY**

Graduate Researcher — ASV Perception and Tracking Fall 2026

- Developed an ASV perception and temporal tracking pipeline integrating YOLO-based detection, eWaSR segmentation, and Unscented Kalman Filter (UKF) state estimation for robust target tracking under dynamic maritime conditions.
- Aligned camera timestamps with vehicle telemetry and integrated RANSAC-based horizon estimation to compensate for camera pitch and roll, enabling monocular range and bearing estimation.
- Evaluated track continuity across 500+ confirmed updates, designing dual-confidence association and soft continuity costs to mitigate low-confidence detections, spatial jumps, and class mismatches; built diagnostic visualizations to isolate data-association failures.
- Achieved successful temporal assignment for 153 of 155 visible target observations with only 2 identity switches in proxy evaluation, and validated track consistency against a manually labeled 300-frame ground-truth reference.

Real-Time IMU-Based Motion Analysis System **New York University, NY**

Embedded Systems Project Mar. 2026 - July 2026

- Implemented a register-level LSM6DSL IMU driver over 400 kHz I2C on an ARM Cortex-M4, configuring sensor parameters and interrupt-driven acquisition for real-time 52 Hz motion sensing.
- Designed a multi-threaded RTOS pipeline with priority scheduling and double buffering to decouple sensor acquisition from CMSIS-DSP FFT processing for continuous frequency-domain analysis on resource-constrained hardware.
- Developed a stateful motion-classification pipeline for tremor, dyskinesia, and Freezing of Gait detection, integrating spectral analysis and motion-state transitions with a custom BLE GATT service for wireless result transmission.

Reinforcement Learning-Based Intelligent Energy Optimization

Independent Research Project Sept. 2025 - Dec. 2025

- Formulated building HVAC control as a Markov Decision Process with defined state, action, and reward spaces, and built an EnergyPlus simulation environment from historical operating data.
- Trained and compared PPO, DDPG, and Q-learning agents over 1,000+ simulation episodes, enabling adaptive temperature control across indoor and outdoor conditions and reducing simulated energy consumption by 20%.

PROFESSIONAL EXPERIENCE

Bolan Intelligent Technology Co., Ltd. **Shenzhen, China**

AI Strategy Intern May 2026 - Aug. 2026

- Built and evaluated AI-agent workflows by translating operational requirements into structured prompts, workflow logic, and evaluation criteria for AI-assisted automation.
- Piloted LLM-powered customer-feedback analysis and recurring reporting workflows; partnered with operations and teams to refine agent behavior, contributing to a 1.6% improvement in operational accuracy.

Financial China Information&Technology Co., Ltd. **Shanghai, China**

Machine Learning Intern Feb. 2025 - Aug. 2025

- Built an XGBoost user-interest prediction model and improved F1 score by 15% through feature/model tuning and grid-search optimization.
- Developed LightGBM-based recommendation logic and validated the production system with A/B testing, increasing conversion rate by 18% and average order value by 10%.

LANGUAGES & SKILLS

- **Programming Languages:** Python, SQL, Java
- **Embedded & Systems:** STM32, Mbed OS, RTOS, CMSIS-DSP, I2C, BLE GATT, PlatformIO
- **AI/ML:** PyTorch, TensorFlow, XGBoost, LightGBM, Reinforcement Learning, LLM/AI-Agent Workflows